

THIRUVENKADAPRASATH.S

192312016

|    |                                                       |
|----|-------------------------------------------------------|
| 13 | Implement the Car Price Prediction Model using Python |
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CSV FILE

EngineSize,Horsepower,Mileage,Price

1.6,105,45000,12000

2.0,150,30000,18500

2.5,190,20000,25000

1.4,90,60000,9500

3.0,250,15000,32000

2.0,140,40000,16000

1.8,125,50000,13500

3.5,300,10000,40000

PROGRAM

```
import numpy as np
```

```
import pandas as pd
```

```
class MultipleLinearRegressionScratch:
```

```
    def __init__(self):
```

```
        self.weights = None
```

```
    def fit(self, X, y):
```

```
        X_b = np.c_[np.ones((len(X), 1)), X]
```

```
        self.weights = np.linalg.inv(X_b.T.dot(X_b)).dot(X_b.T).dot(y)
```

```

def predict(self, X):
    X_b = np.c_[np.ones((len(X), 1)), X]
    return X_b.dot(self.weights)

df = pd.read_csv("car_price_data.csv")

X = df[["EngineSize", "Horsepower", "Mileage"]].values
y = df["Price"].values

model = MultipleLinearRegressionScratch()
model.fit(X, y)
predictions = model.predict(X)

mse = np.mean((y - predictions) ** 2)
r2 = 1 - (np.sum((y - predictions) ** 2) / np.sum((y - np.mean(y)) ** 2))

print(f"Intercept (Bias): {model.weights[0]:.2f}")
print(f"Coefficients (Weights): {model.weights[1:]}")
print(f"Mean Squared Error: {mse:.2f}")
print(f"R2 Score: {r2:.4f}")

sample_car = np.array([[2.0, 145, 35000]])
predicted_price = model.predict(sample_car)[0]

print(f"\nPredicted Price for car [2.0L, 145 HP, 35,000 miles]: ${predicted_price:.2f}")

```

OUTPUT

File Edit Selection View Go Run ...

CandidateElimination

EXP13.py

car\_price\_data.csv

EXP13.py

CandidateElimination

Outline

Timeline

EXP13.py > ...

27

28 predictions \*\* 2)

29 / - predictions) \*\* 2) / np.sum((y - np.mean(y)) \*\* 2))

30

31 'Bias): (model.weights[0]:.2f)')

32 :s (Weights): (model.weights[1:]')

33 d Error: (mse:.2f)')

34 'r2:.4f)')

35

36 'ay([[2.0, 145, 35000]])

37 model.predict(sample\_car)[0]

38

39 | Price for car [2.0L, 145 HP, 35,000 miles]: \${predicted\_price:.2f})'

Problems Output Debug Console Terminal Ports

Python +

Python

PS C:\Users\student\Desktop\CandidateElimination> & C:\Users\student\AppData\Local\Programs\Python\Python312\python.exe c:/Users/student/Desktop/CandidateElimination/EXP13.py

Intercept (Bias): 2453.06

Coefficients (Weights): [-2.34354662e+03 1.53148210e+02 -6.72578889e-02]

Mean Squared Error: 265140.07

R2 Score: 0.9973

Predicted Price for car [2.0L, 145 HP, 35,000 miles]: \$17618.46

PS C:\Users\student\Desktop\CandidateElimination>

Chat

Sessions

Tip: Use /create-agent to scaffold a custom agent for your workflow.

+ EXP13.py

Describe what to build

+ Agent Auto

Activate Windows

Go to Settings to activate Windows.

Spaces: 4 UTF-8 Python Python 3.12 (64-bit)